

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lectures 11 (Ch. 6)

Extension of the IS-LM

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
 1. Basic
 2. Extended
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

IS LM MODEL

- Fixed P
- Only 1 bond and 1 interest rate
- CB Policy rate = borrowing cost

REALITY

- Prices change over time
- Many financial assets with different risk level
=> different interest rates

EXTENDED IS-LM model to have a **more realistic setting**

Taking into account **two elements**

1. Change in prices (π) => variables can be expressed in nominal and real terms
 r = real interest rate
1. Interest rate controlled by CB is not exactly the borrowing rate => since borrowing money implies a risk, banks charge an higher interest rate that depends on risk premium (x)
 $r+x$ = borrowing rate

REAL VS NOMINAL INTEREST RATE

Distinguish btw the two is relevant if there is inflation in the economy

Problem: Suppose you want a coffee that costs 1€ but you don't have any money with you. What should you do?

Ask for a loan to a friend!

What you can do in the future?

1) Pay your debt with money

How much you will have to *pay back* in the future depends on ***nominal interest rate***

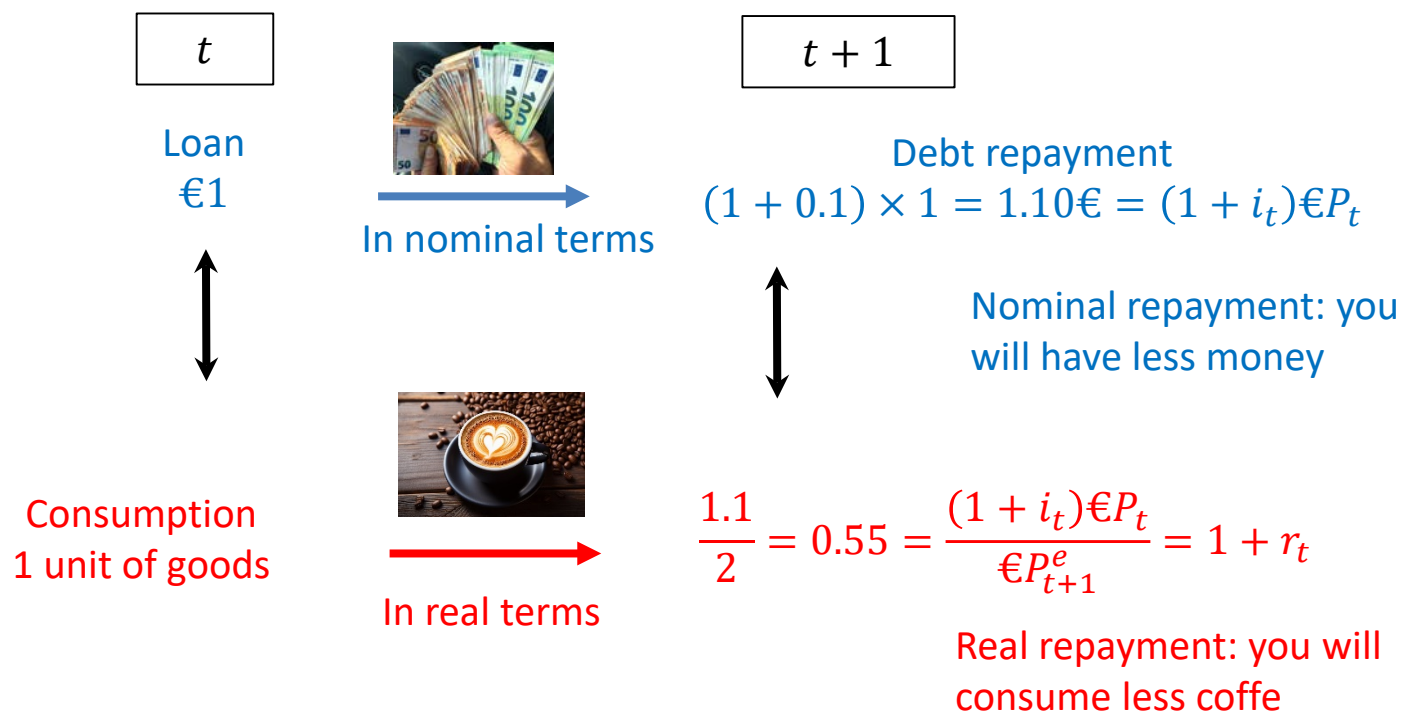
2) Pay your debt with coffees

How many coffees you have to give to your friend depend on today and tomorrow **prices** and the **nominal interest rate**

$i = 10\%$ = nominal interest rate

$€P_t = 1 €$ = current price of 1 unit of goods

$€P_{t+1}^e = 2€$ = expected price for the next period



$$1 + r_t = -0.55 \Rightarrow r_t = -0.45$$

Same reasoning if today you do not take a coffee today

⇒ You save 1€

⇒ By investing this 1€, in one year from now you will have 1,1€ (nominal return)

⇒ In the next period you can take $1,1/2=0,5$ coffee more (real return)

RELATION BTW REAL AND NOMINAL INTEREST RATE

$$1 + r_t = (1 + i_t) \frac{\text{€}P_t}{\text{€}P_{t+1}^e}$$

If at time t we want to consume 1 unit more (less) of goods, we need P_t euros more (less)

⇒ at $t+1$ we repay (obtain) $(1 + i_t)$ unit of money

⇒ we can consume $(1 + i_t) \frac{\text{€}P_t}{\text{€}P_{t+1}^e}$ more (less) units of goods

Nominal interest rate (i_t): Amount of **euro** we must pay in the future to get an additional **euro** today

If at t we save (borrow) 1€, at $t+1$ we get (have to pay) $(1+i_t)$ €

Real interest rate (r_t): Amount of **goods** we must give in the future in exchange of a **good** today

If at t we save (borrow) the amount necessary to buy 1 unit of goods that costs P_t , at $t+1$ we get (have to pay) the equivalent of $(1+r_t)$ units of good that will cost P_{t+1}

The two interest rates are linked through the **dynamic of prices btw t and t+1**

$$(1 + i_t) \left(\frac{\epsilon P_t}{\epsilon P_{t+1}^e} \right) = 1 + r_t$$

We can simplify this expression to better understand the link using the inflation rate concept in a given period

$$\pi_{t+1} = \frac{(\epsilon P_{t+1} - \epsilon P_t)}{\epsilon P_t}$$

...But, as future prices are not known but simply expected, we have the **expected inflation rate**:

$$\pi_{t+1}^e = \frac{(\epsilon P_{t+1}^e - \epsilon P_t)}{\epsilon P_t}$$

Rearranging

$$\pi_{t+1}^e = \frac{\epsilon P_{t+1}^e}{\epsilon P_t} - 1 \quad 1 + \pi_{t+1}^e = \frac{\epsilon P_{t+1}^e}{\epsilon P_t} \quad \left(\frac{\epsilon P_t}{\epsilon P_{t+1}^e} \right) = \frac{1}{1 + \pi_{t+1}^e}$$

And substituting the price ratio in terms of inflation:

$$(1 + i_t) \frac{1}{1 + \pi_{t+1}^e} = 1 + r_t$$

Exact relationship btw. nominal and real interest rates:

$$\frac{1 + i_t}{1 + \pi_{t+1}^e} = 1 + r_t$$

For small values, we can use an approximation:

$$r_t \approx i_t - \pi_{t+1}^e$$


$i_t \uparrow$ also $r_t \uparrow$:

if I save I save and the nominal interest rate increases, tomorrow I will be able to consume more goods, given their expected prices.

If I borrow today and the nominal interest rate increases, tomorrow I will have to give up more goods, given their expected prices.

$\pi_{t+1}^e \uparrow$ hence $r_t \downarrow$:

If I save today and the price level increases, tomorrow I will be able to consume fewer goods that will cost more

If I borrow today, tomorrow I will have to give up fewer units of goods due to higher prices

Lenders are better off when $i \uparrow$ and $\pi_{t+1}^e \downarrow$

Borrowers are better off when $i \downarrow$ and $\pi_{t+1}^e \uparrow$

OBSERVATIONS

$$r_t \approx i_t - \pi_{t+1}^e \quad \leftarrow$$

The CB with MP changes i in order to reach a specific value of r , given our expectations

- If people expect constant prices (short run):

$$\epsilon P_{t+1}^e = \epsilon P_t \Leftrightarrow \pi_{t+1}^e = 0 \Leftrightarrow r_t = i_t$$

- When we expect increase in prices (normal situation for a healthy economy):

$$\epsilon P_{t+1}^e > \epsilon P_t \Leftrightarrow \pi_{t+1}^e > 0 \Leftrightarrow r_t < i_t$$

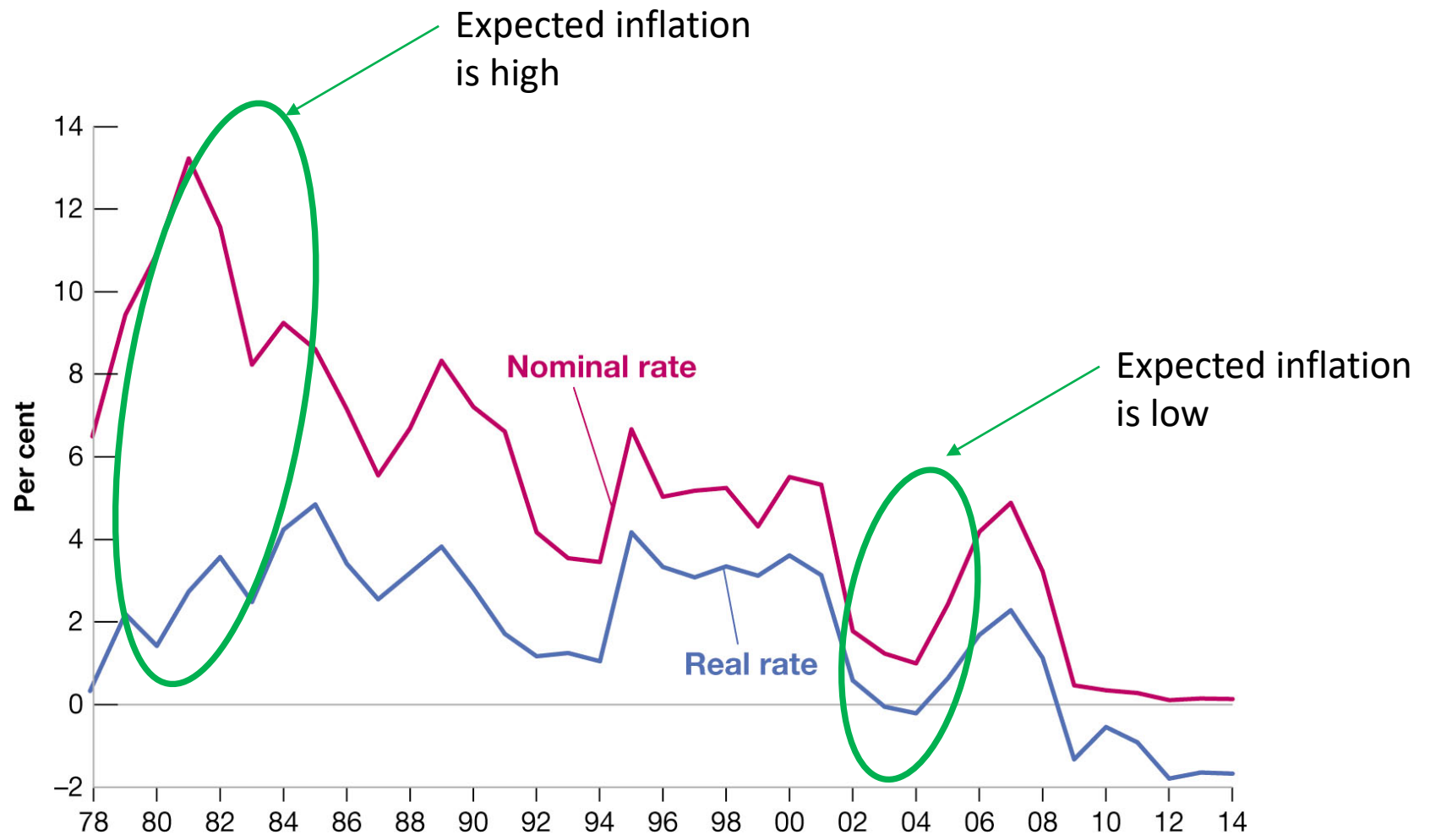
- When we expect decrease in prices (deflation):

$$\epsilon P_{t+1}^e < \epsilon P_t \Leftrightarrow \pi_{t+1}^e < 0 \Leftrightarrow r_t > i_t$$

- In the liquidity trap (at the zero-lower bound) the CB sets $i = 0 \Rightarrow r_t \approx -\pi^e$

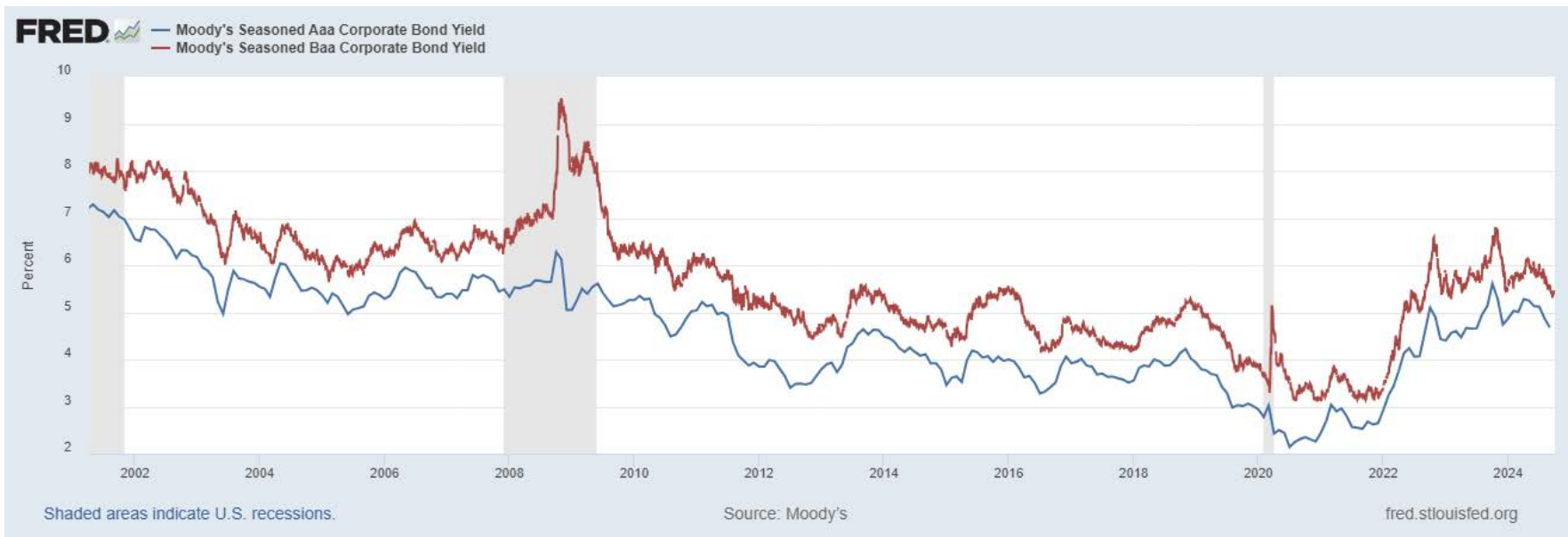
=> The real interest rate depends on our expectations

- If we expect inflation : $r_t < 0$
- If we expect deflation : $r_t > 0$



RISK PREMIUM

In reality there is a probability that a borrower will not repay => lenders ask for a risk premium



There is some probability that the borrower does not repay (**default risk**) :
to be compensated for the risk bonds' holders require a **risk premium**



Policy interest rate $\neq$ **borrowing rate**

chosen by CB (i)

charged by bank to firms/HH $> i$

Risk free bonds

- Guaranteed repayment
- Return = i

Risky bonds

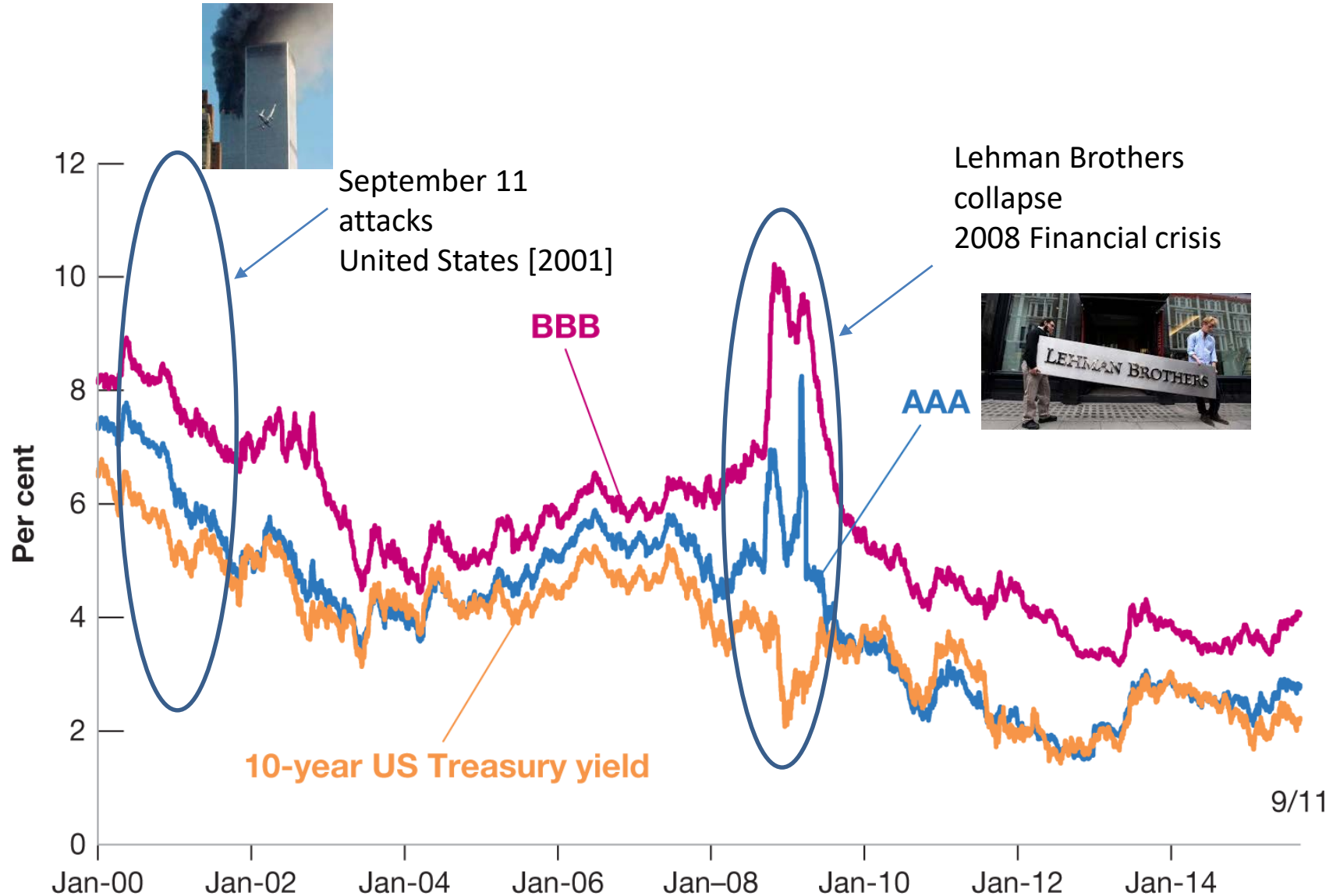
- Repayment uncertain
- Lenders ask for a higher interest rate
 $= i + x$

 Risk premium/spread
the additional amount
required by investors to buy a
risky asset

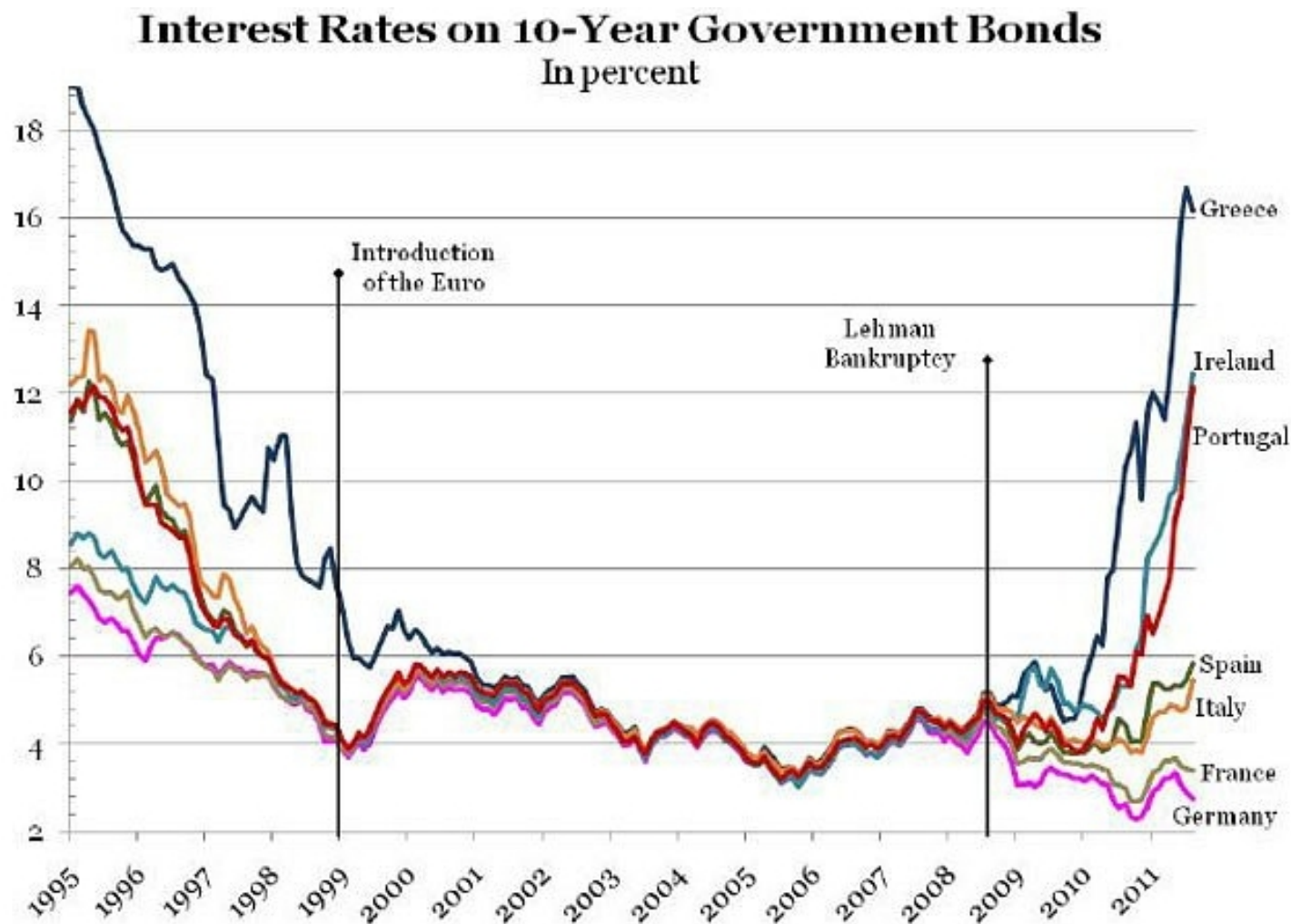
Risk premium/spread depends on

- Lenders' risk aversion (preferences) :
if they are more risk averse, they demand higher return to lend money
- default probability (structural features, business cycles, time horizon captured by assets' rating):
as the probability of default increases, as lenders demand a higher return

Return on 10-year US bonds and corporate bonds AAA and BBB: there is a risk premium associated to risky assets
Uncertainty increases the risk premium



The euro adoption reduced uncertainty, while the financial crisis increased it



EXTENDED IS-LM

Novelty: expenditure decisions depend on **real interest rate** adjusted for the **risk**



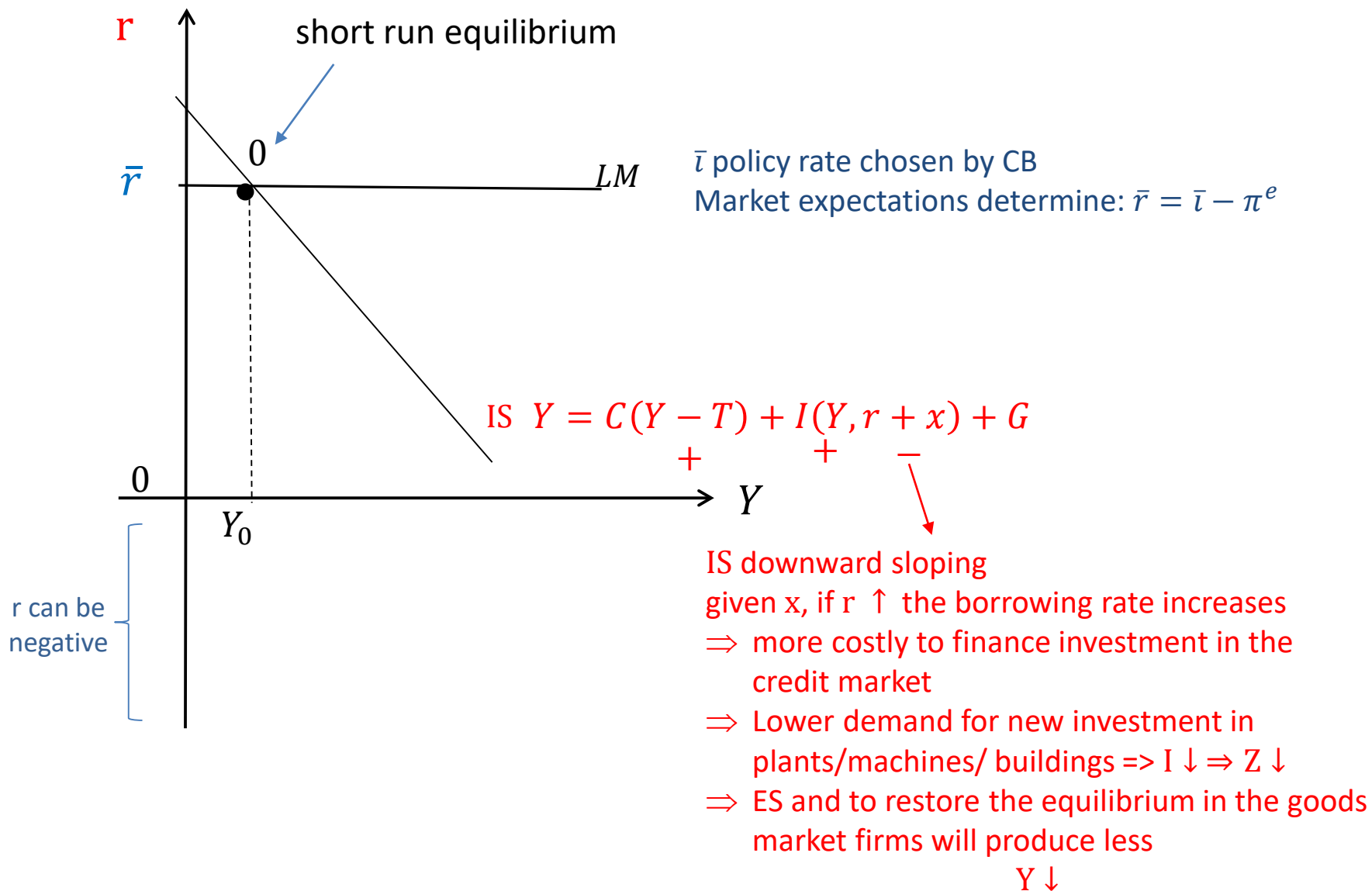
Relevant interest rate for expenditure decisions in C and I is the

$$\text{Borrowing rate: } r + x = i - \pi^e + x$$

CB sets the NOMINAL interest rate (the cost of money): i and r is determined indirectly based on expectations

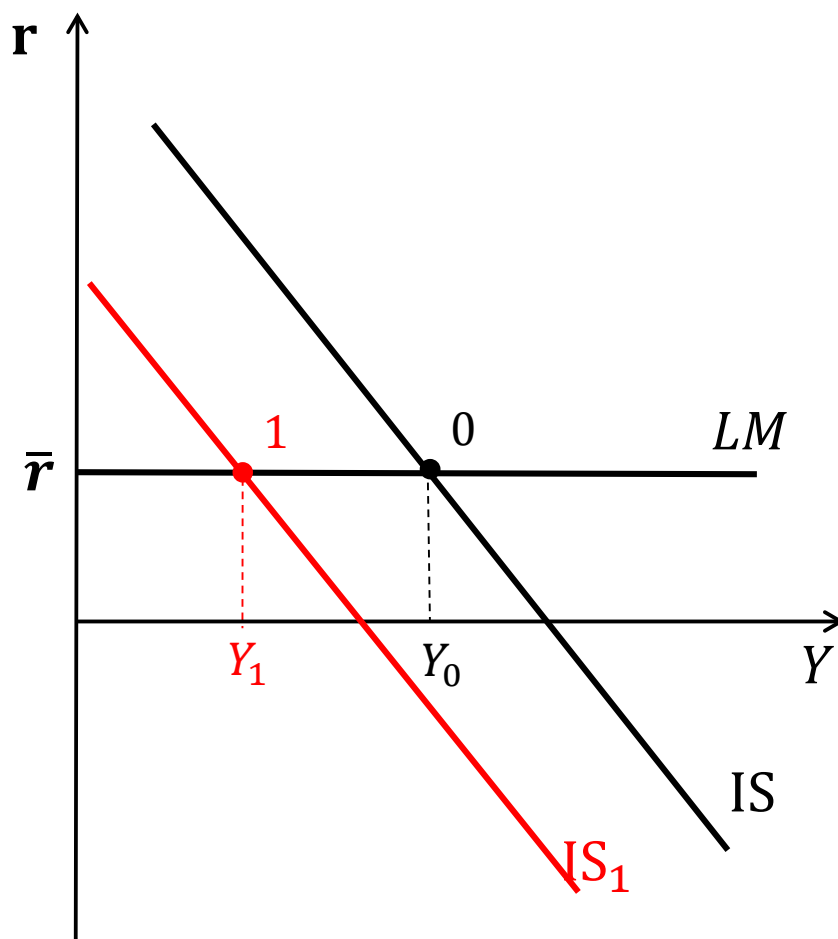
Agents and firms: have expectations about prices and interested in how many goods they will have to give up in the future to get an additional one today

Graph: in the (Y,r) plane



A SHOCK in FINANCIAL MARKETS: doubts about the solvency of borrowers as during the 2008 financial crisis after the collapse of Lehman Brothers

Higher risk aversion: $x \uparrow$



Which curve is affected?

IS to the LEFT: $x \uparrow \Rightarrow I \downarrow \Rightarrow Y \downarrow$

$LM =$

Equilibrium moves from 0 to 1

What happens economically?

The higher borrowing rate makes more costly to finance investment in the credit market, reduces investment, aggregate demand, and through the multiplier production in the economy

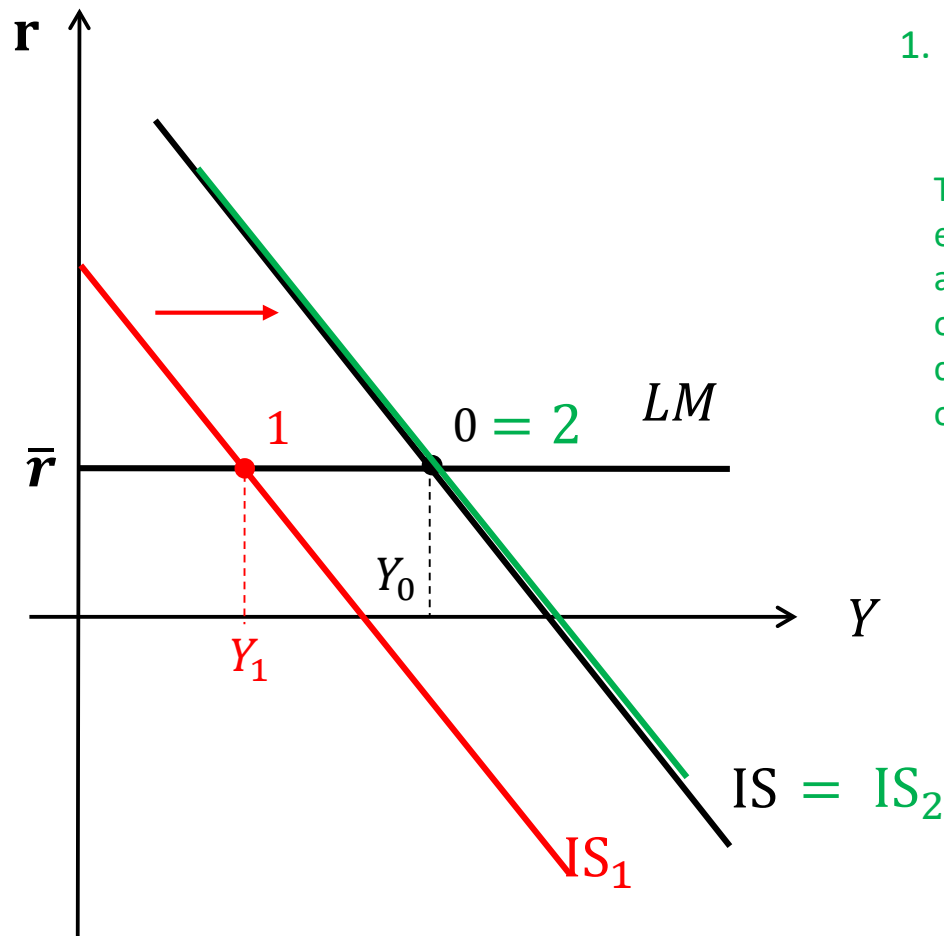
Does the CB do anything?

Yes also if the LM curve does not change since $Y \downarrow \Rightarrow M^D \downarrow \Rightarrow i \downarrow \& r \downarrow$

To avoid the change in the interest rate, the CB modifies the monetary base to $\downarrow M^S$

$\Rightarrow i$ e r are unchanged

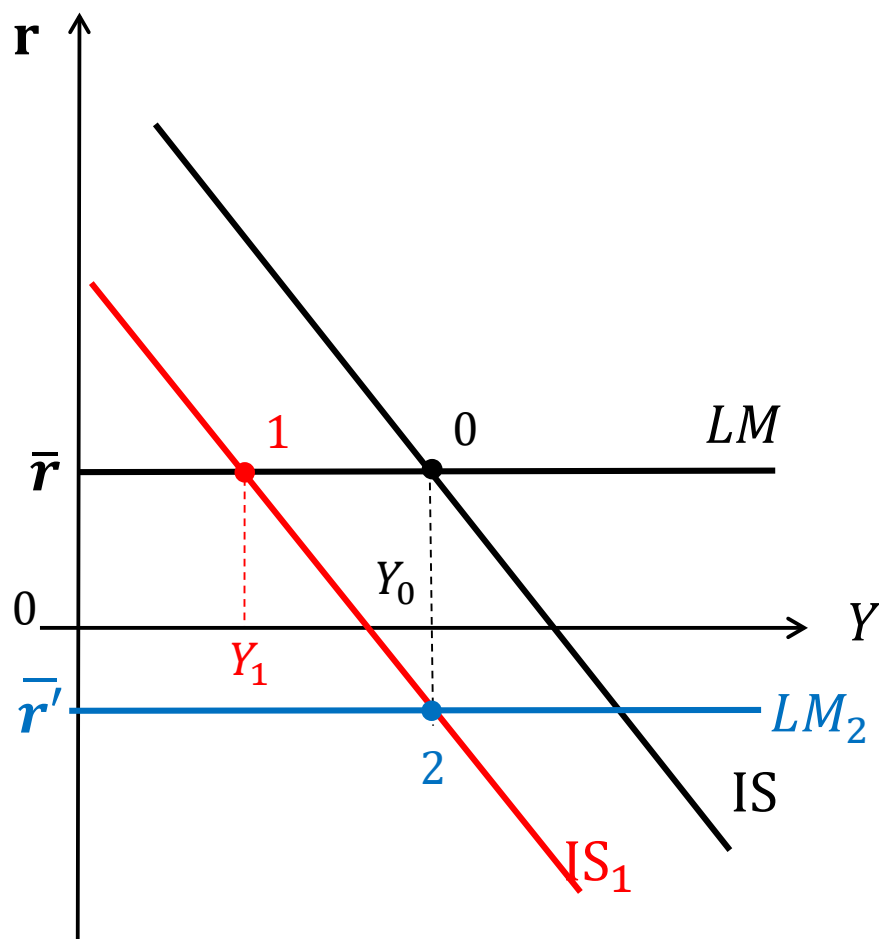
Which economic policies to contrast the recession?



1. Expansionary FP : IS to the right up to the initial position

The government could increase public expenditure or reduce net taxes to stimulate aggregate demand and, in equilibrium, the level of production. In principle, this policy could completely offset the initial negative effect on output.

Which economic policies to contrast the recession?



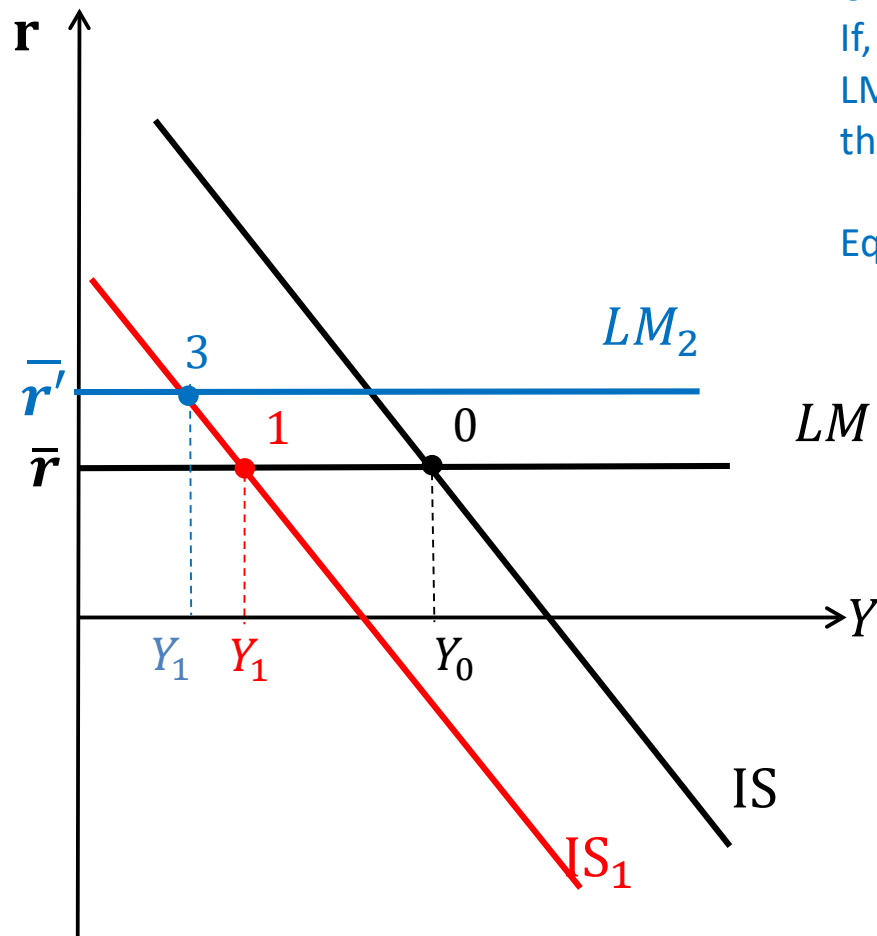
2. Expansionary MP: LM down

The CB buys bonds $\Rightarrow \epsilon P_B \uparrow \Rightarrow i \downarrow$,
and given expected inflation also $r \downarrow$

But in this graph to come back at the
initial level of output r must be
negative $\Rightarrow$ agents need to expect
higher prices

The central bank's actions alone may
not be sufficient to ensure a negative
real interest rate.

The problems faced by the CB are particularly evident if before the shock $i = 0$



CB no longer able to reduce $\bar{r}$

If, in addition, $\pi^e < 0 \Rightarrow r$ increases

LM curve moves up without any intervention if the CB and output further decrease

Equilibrium moves from 1 to 3

QUICK CHECKS

If expected inflation is -2% and the nominal interest rate is 3%, then the real interest rate is (approximately) equal to

- a) **5 %**
- b) 1 %
- c) - 1 %
- d) - 5%

QUICK CHECKS

If as a result of an energy crisis agents expect higher prices in the future, to offset the effect on the country's output level the CB has to

- a) reduce the nominal interest rate
- b) increase the nominal interest**
- c) increase public expenditure
- d) take constant the money supply

CONCLUSIONS

- More realistic model
- Now we can use the model to analyze new scenarios
 - A crisis on the financial markets
 - Problems in the transmission mechanism of the monetary policy
 - ...

TAKE AWAY AND KEY CONCEPTS

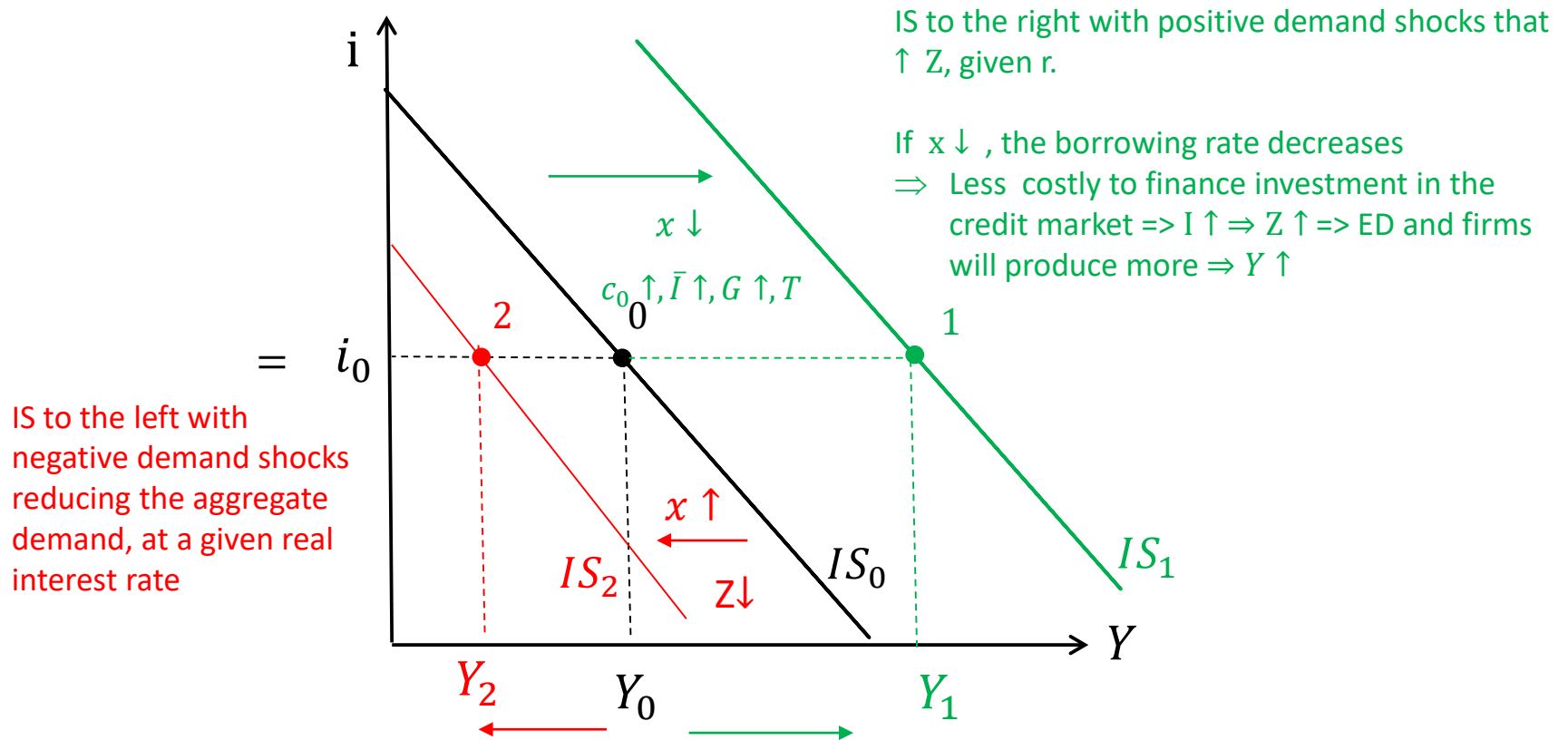
- Real interest rate
 - Relationship (exact and approximated) btw nominal and real interest rate
- Risk premium
- Borrowing rate
- Policy rate vs. real interest rate in the LM
- IS position
- LM position
- Negative real interest rate

LET'S MAKE A SURVIVAL RECAP

NOVELTY

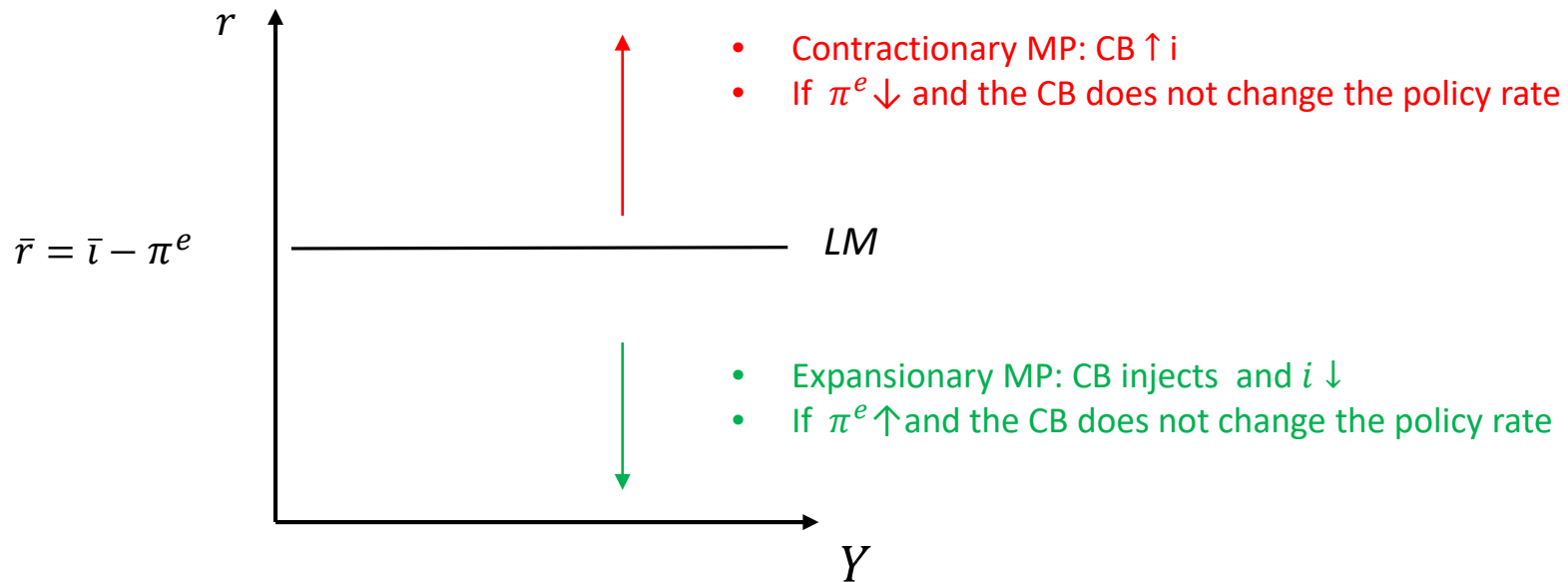
The position of the IS depends on exogenous variables influencing the output level in the goods market, given r

Also x is an exogenous variable for the IS



NOVELTY

Also x is an exogenous variable for the LM π^e



**FOR THOSE WHO WANT TO GO MORE
IN DEPTH... BUT NOT COMPULSORY**

RISK AVERSION

- $i \equiv$ interest rate paid by a risk-free asset

Return paid by a risk-free asset = $(1 + i)$

- $p \equiv$ **probability of default of the borrower**
- $x \equiv$ **Risk premium**

The expected return of a risky asset will be

$$EV = p \times 0 + (1 - p)(1 + i + x)$$

Arbitrage condition in financial markets:

In order to exchange both risky and risk – free assets the expected return of the risky asset must be equal to the return paid by the risk-free asset

– If $(1 + i) > (1 - p)(1 + i + x) \Rightarrow$ Investors buy risk-free assets only

– If $(1 + i) < (1 - p)(1 + i + x) \Rightarrow$ Investors buy risky assets only

– Both assets are exchanged if:

$$(1 + i) = (1 - p)(1 + i + x)$$

Given the arbitrage condition

$$(1 + i) = (1 - p)(1 + i + x)$$

We can solve for the risk premium: x

$$(1 + i) = 1 + i + x - p - pi - px$$

$$x - px = p(1 + i)$$

$$x(1 - p) = p(1 + i)$$

$$x = \frac{p}{1 - p} (1 + i)$$

- If $p \uparrow \Rightarrow$ higher default probability $\Rightarrow x \uparrow$
- If investors were risk averse, x would be higher